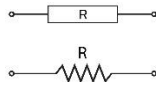


RESISTOR

A resistor is a passive electronic component made of resistive materials (like carbon film or metal film or wire wound or etc.,). It opposes the flow of electric current.

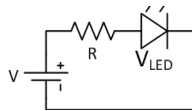


When current flows through a resistor electrical energy is converted into heat due to collisions of electrons within the resistive material.

Need of a resistor in an electronic circuit

Assume, we connect a 5V LED to 10V power supply and the current is 20mA.

In this case LED needs only 5V, if we connect it directly to the 10V then the excessive current will damage the LED. We can add resistor to drop 5V.



Voltage across the resistor is Voltage across the power supply – Voltage across the LED.

$$V_r = 10V - 5V = 5V$$

$$\text{Value of resistor} = V_r / I = 5 / 0.02 = 250 \text{ Ohm.}$$

Types of resistors

Based on the requirement resistors are divided as,

Fixed resistors

This type of resistors provides a non adjustable constant resistance value in them throughout service life

Carbon film resistor

Ceramic base contains a deposition of thin layer of carbon. Resistance value was adjusted by trimming the layer of carbon film. It provides a moderate stability and accuracy and less precision.



Metal film resistor

In this resistor resistive path made up on ceramic core by applying thin metal layer. It provides a long term stability and higher accuracy than carbon film resistors.



Wire wound resistor

These resistors are constructed by winding ceramic metal wire around an insulated core. Resistance value depends on the wire length, material and thickness.



Thermistor

These types of resistors are made of semiconductor material and the resistance value were change based on the surrounding temperature.



Negative temperature coefficient (NTC): resistance decreases when temperature rises.

Positive temperature coefficient (PTC): resistance increases when temperature rises.

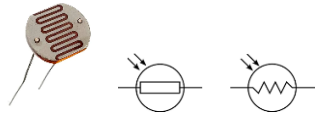
Varistor

These types of resistors are changes their resistance values based on the voltage applied across it. This will help to protect circuits from dangerous voltage.



Photoresistor

These types of resistors resistance values are decreases when the intensity of incident light increases. It's also known as LDR.



Variable resistors

If the circuit needs an adjustable resistance control then we can choose the variable resistors like rheostat and potentiometer.

How resistors are used in PCBs,

Through hole resistor: It contains wire leads that pass through PCB holes and soldered on the opposite side.

SMD resistor: these types of resistors are made upon film based technology. This is directly soldered onto the PCB surface through. These resistors are very small in size.



Package of surface mount resistor

Package size	dimensions
0805	.08 x .05 inch
0603	.06 x .03 inch

How to find the value of resistor,

Color coding – Band color and Value

Black – 0	Green – 5
Brown – 1	Blue – 6
Red – 2	Violet – 7
Orange – 3	Gray – 8
Yellow – 4	White – 9

For tolerance

Silver - $\pm 10\%$	Gold - $\pm 5\%$
---------------------	------------------

4 Band resistors

1,2 band – digit	3 band – multiplier
4 band – tolerance	

Example: Brown Brown Red Gold

$$11 \times 10^2 \pm 5\% = 1100 \pm 5\% \text{ Ohm.}$$

5 band resistors

1,2,3 band – digit	4 band – multiplier
5 – tolerance	

Example: Red Violet Black Black Gold

$$270 \times 10^0 \pm 5\% = 270 \pm 5\% \text{ Ohm}$$

SMD resistor code

1 st & 2 nd digit – value	3 rd digit – multiplier
---	------------------------------------

Example,

$$103 = 10 \times 10^3 \text{ Ohm}$$

$$1001 = 100 \times 10^1 \text{ Ohm}$$

$$100 = 10 \times 1 \text{ Ohm}$$

$$4R7 = 4.7 \text{ Ohm}$$

$$R10 = .1 \text{ Ohm}$$

Equivalent resistance based on connection

In series: $R_{eq} = R1 + R2 + \dots + Rn$

In parallel: $1/R_{eq} = 1/R1 + 1/R2 + \dots + 1/Rn$

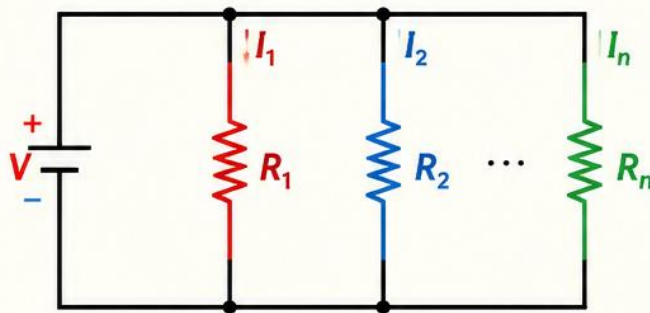
In series, current has only one path so the same current flows through all resistors that's why we add all resistance to get the equivalent resistance. Therefore the total resistance increases.

In parallel, current has multiple path and divides among branches. Voltage is the same across all branches. Therefore the total resistance decreases.

WHY $1/R$ IN PARALLEL CONNECTION?

In a parallel connection, the **reciprocal of equivalent resistance** equals the **sum of reciprocals** of individual resistances.

1 PARALLEL CIRCUIT



✓ All resistors are connected across the same two points.

✓ Therefore, voltage across each resistor is the same.

$$V_1 = V_2 = \dots = V_n = V$$

(Constant)

2 OHM'S LAW

$$V = IR \quad (\text{Ohm's Law})$$

Current in each branch:

$$I = \frac{V}{R}$$

So, for each resistor:

$$I_1 = \frac{V}{R_1} \quad I_2 = \frac{V}{R_2} \quad \dots \quad I_n = \frac{V}{R_n}$$

3 TOTAL CURRENT

In parallel connection, total current (I) is the sum of currents in all branches.

$$I = I_1 + I_2 + \dots + I_n$$

$$I = \frac{V}{R_1} + \frac{V}{R_2} + \dots + \frac{V}{R_n}$$

$$I = V \left[\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n} \right]$$

$$I = V \left[\frac{1}{R_{eq}} \right]$$



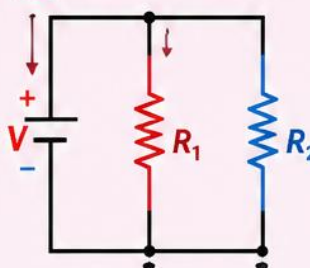
Voltage (V) is common in all branches, so it can be taken out as a common factor.

4 THEREFORE,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

The reciprocal of the equivalent resistance is equal to the sum of reciprocals of individual resistances.

5 EXAMPLE (2 RESISTORS)



Equivalent resistance:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

or ↓

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$



KEY TAKEAWAYS

✓ In parallel, voltage is same across all branches.

✓ Currents add up, but voltage stays constant.

✓ That's why $1/R$ appears in the formula!



RESISTOR

Applications

- Current limiting
- Voltage division
- Heating elements
- Volume and speed control

Application in electronic Circuit

- Voltage divider
- Signal filtering
- Signal turning
- Pull up and pull down resistors
- Transistor biasing
- Timing circuit